
CS771: Introduction to Machine Learning

Minor Assignment 2 Report

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Abstract

This report investigates simultaneous classification and imputation of missing pixels using a class-conditional Gaussian generative model. We mathematically derive a single-step joint MAP inference procedure and implement it by modifying the test-time pipeline. We then empirically compare this single-step approach against standard two-step inference using the MNIST dataset under varying synthetic censoring levels (2% to 51%). Analyzing accuracy, runtime, and reconstruction quality, we conclude that while single-step inference rigorously models uncertainty, its practical performance heavily depends on the proportion of observed data.

1 Question 1: Inference with Missing Pixels

1.1 Single-Step Inference Derivation

1. Joint Objective Formulation:

We aim to compute the Maximum A Posteriori (MAP) estimate for both the class c and the missing pixels x_m :

$$(\hat{y}, \hat{x}_m) = \arg \max_{c, v} P(y = c, x_m = v \mid x_o) \quad (1)$$

Using the chain rule of probability:

$$P(y = c, x_m = v \mid x_o) = P(y = c \mid x_o) P(x_m = v \mid y = c, x_o) \quad (2)$$

2. Inner Maximization (Solving for $\hat{x}_m$):

For a fixed class c , we maximize:

$$\max_v P(x_m = v \mid y = c, x_o) \quad (3)$$

We model the full data vector as $x = [x_o, x_m]$ following a multivariate Gaussian:

$$x \mid y = c \sim \mathcal{N}(\mu_c, \Sigma_c) \quad (4)$$

Partition the mean and covariance as:

$$\mu_c = \begin{bmatrix} \mu_{o,c} \\ \mu_{m,c} \end{bmatrix}, \quad \Sigma_c = \begin{bmatrix} \Sigma_{oo,c} & \Sigma_{om,c} \\ \Sigma_{mo,c} & \Sigma_{mm,c} \end{bmatrix}$$

Using properties of multivariate Gaussians, the conditional distribution is:

$$P(x_m | x_o, y = c) = \mathcal{N}(\bar{\mu}_c, \bar{\Sigma}_c) \quad (5)$$

where

$$\bar{\mu}_c = \mu_{m,c} + \Sigma_{mo,c} \Sigma_{oo,c}^{-1} (x_o - \mu_{o,c}) \quad (6)$$

$$\bar{\Sigma}_c = \Sigma_{mm,c} - \Sigma_{mo,c} \Sigma_{oo,c}^{-1} \Sigma_{om,c} \quad (7)$$

Since a Gaussian distribution is maximized at its mean, the optimal estimate of the missing pixels is:

$$\hat{x}_m = \bar{\mu}_c \quad (8)$$

The maximum value of the conditional density is:

$$\max_v P(x_m = v | y = c, x_o) = \frac{1}{(2\pi)^{d/2} |\bar{\Sigma}_c|^{1/2}} \quad (9)$$

3. Outer Maximization (Solving for $\hat{y}$):

Substituting back into the joint objective:

$$\hat{y}_{\text{single}} = \arg \max_c \left[P(y = c | x_o) \frac{1}{(2\pi)^{d/2} |\bar{\Sigma}_c|^{1/2}} \right] \quad (10)$$

Since $(2\pi)^{d/2}$ is constant across classes, it can be ignored:

$$\hat{y}_{\text{single}} = \arg \max_c \frac{P(y = c | x_o)}{\sqrt{|\bar{\Sigma}_c|}} \quad (11)$$

This formulation efficiently reduces the joint optimization into a closed-form solution by exploiting properties of conditional Gaussians, avoiding any iterative procedures.

1.2 Comparison of Inference Procedures

Are the single-step and two-step inference procedures identical?

Answer: No, they are not identical.

Mathematical Justification:

The two-step inference procedure predicts the class as:

$$\hat{y}_{\text{two}} = \arg \max_c P(y = c | x_o) \quad (12)$$

In contrast, the single-step procedure modifies this by incorporating uncertainty in the missing pixels:

$$\hat{y}_{\text{single}} = \arg \max_c \left[P(y = c | x_o) |\bar{\Sigma}_c|^{-1/2} \right] \quad (13)$$

The determinant $|\bar{\Sigma}_c|$ represents the conditional uncertainty (or volume) of the missing pixels given the observed pixels.

Thus, the single-step method penalizes classes with higher uncertainty in reconstructing missing pixels, whereas the two-step method ignores this factor entirely.

Conclusion. The two procedures coincide only in the special case where $|\bar{\Sigma}_c|$ is identical for all classes. In general, however, they produce different decision boundaries, and hence are not equivalent.

2 Question 2: Code Modifications and Pseudo-code for the Single-Step Technique

2.1 Objective

The objective of this question is to explain how the lecture code should be modified to implement the single-step technique without changing the training procedure. The Gaussian model learned during training remains unchanged. Only the testing or inference stage is reorganized so that classification and reconstruction are performed together.

2.2 Existing Lecture Code

In the lecture notebook, the relevant functions are:

- `learnClassConditionalDist(...)`: learns the mean μ , covariance Σ , and prior probability p for each class.
- `predictClassScores(...)`: computes the score of one class using the observed pixels.
- `predictGen(...)`: evaluates all classes and selects the class with the maximum score.
- `reconstructImages(...)`: reconstructs the missing pixels after classification.

Thus, the current inference pipeline is:

1. Train the class-wise Gaussian model.
2. Compute class scores using the observed pixels.
3. Select the predicted class using the maximum score.
4. Reconstruct missing pixels separately.

2.3 Main Modification Required

The single-step technique changes only the inference stage. Instead of first predicting the class label and then reconstructing the missing pixels in a separate function, both tasks are merged into one unified inference routine.

For each test image and for each class:

- compute the observed-data score,
- compute the conditional reconstruction of missing pixels,
- retain the class with the highest score,
- output both the predicted class and the corresponding reconstruction.

Therefore, the single-step method is only a reorganization of test-time computation and does not change the training equations.

2.4 Summary Table

2.5 Mathematical Formulation

Let O denote the set of observed pixel indices and M denote the set of missing pixel indices.

For each class c , the observed-data score is computed as

$$\text{score}_c = \log p(c) - \frac{1}{2} [\log \det(\Sigma_c) + (x_O - \mu_c[O])^T \Sigma_c[O, O]^{-1} (x_O - \mu_c[O])].$$

The missing pixels are reconstructed using the Gaussian conditional mean:

$$\hat{x}_M = \mu_c[M] + \Sigma_c[M, O] \Sigma_c[O, O]^{-1} (x_O - \mu_c[O]).$$

The final predicted class is

$$c^* = \arg \max_c \text{score}_c.$$

The reconstruction corresponding to class c^* is then used as the final reconstructed image.

Table 1: Summary of code modifications required for the single-step technique

Code Component	Current Role	Change for Single-Step Method
<code>learnClassConditionalDist(...)</code>	Learns the class-wise Gaussian parameters μ , Σ , and prior p	No change; training remains exactly the same.
Data loading, normalization, and flattening	Prepares the input images for training and testing	No change.
Mask creation	Identifies observed and missing pixels	No change.
<code>predictClassScores(...)</code>	Computes the score of a single class using observed pixels	Its logic is reused inside the merged inference function.
<code>predictGen(...)</code>	Loops over classes and returns the predicted label	Removed as a separate stage; its class loop becomes part of the single-step routine.
<code>reconstructImages(...)</code>	Reconstructs missing pixels after classification	Its reconstruction formula is merged into the same inference routine.
New merged function	Not present in the lecture code	Introduce a function such as <code>SingleStepPredictAndReconstruct(...)</code> that performs scoring and reconstruction together.

Note on constants. Constant terms are omitted as they do not influence the arg max selection.

2.6 Implementation Notes

There are three important implementation details:

- If O and M are stored as index lists in NumPy, then `Sigma[0, 0]` does not produce the required submatrix. The correct form is `Sigma[np.ix_(0, 0)]` and `Sigma[np.ix_(M, 0)]`.
- Since the covariance submatrix may be singular, the pseudoinverse `pinv(...)` should be used instead of the standard inverse.
- For numerical stability, `slogdet(...)` returns `(sign, logdet)`. Since a valid covariance determinant should be positive, the `sign` must be checked and `logdet` is then used in the score computation.

2.7 Pseudo-code

`SingleStepPredictAndReconstruct(X, model, mask):`

```

O = observed indices
M = missing indices

for each sample i:
    bestScore = -infinity
    bestClass = None
    bestRecon = None

    for each class entry in model:
        (mu, Sigma, c, p) = class entry

        x_o = X[i, O]
        mu_o = mu[O]
        mu_m = mu[M]

        S_oo = Sigma[np.ix_(O, O)]

```

```

S_mo = Sigma[np.ix_(M, 0)]

invS = pinv(S_oo)

(sign, logdet) = slogdet(Sigma)

if sign <= 0:
    stableLogDet = 0
else:
    stableLogDet = logdet

score = log(p) - 0.5 * (
    stableLogDet +
    (x_o - mu_o)^T invS (x_o - mu_o)
)

recon_m = mu_m + S_mo @ invS @ (x_o - mu_o)

if score > bestScore:
    bestScore = score
    bestClass = c
    bestRecon = recon_m

yPred[i] = bestClass
XRecon[i, 0] = X[i, 0]
XRecon[i, M] = bestRecon

return yPred, XRecon

```

2.8 Conclusion

In conclusion, the single-step technique does not modify the training procedure. The Gaussian parameters μ , Σ , and prior probabilities learned in the lecture notebook remain unchanged. Only the inference stage is reorganized. Instead of separately classifying the image and then reconstructing the missing pixels, the new merged inference routine evaluates each class once, computes both the observed-data score and the missing-pixel conditional estimate, and returns the class with the highest score together with its reconstruction. Hence, the single-step technique is fully consistent with the mathematical derivation obtained by Pair 1.

2.9 Figures

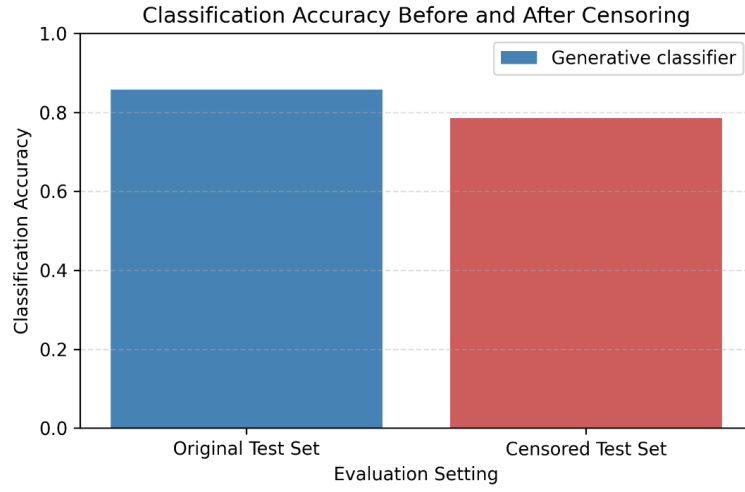


Figure 1: Classification accuracy of the generative classifier on the original test set and on censored test images. Accuracy decreases after censoring, as expected.

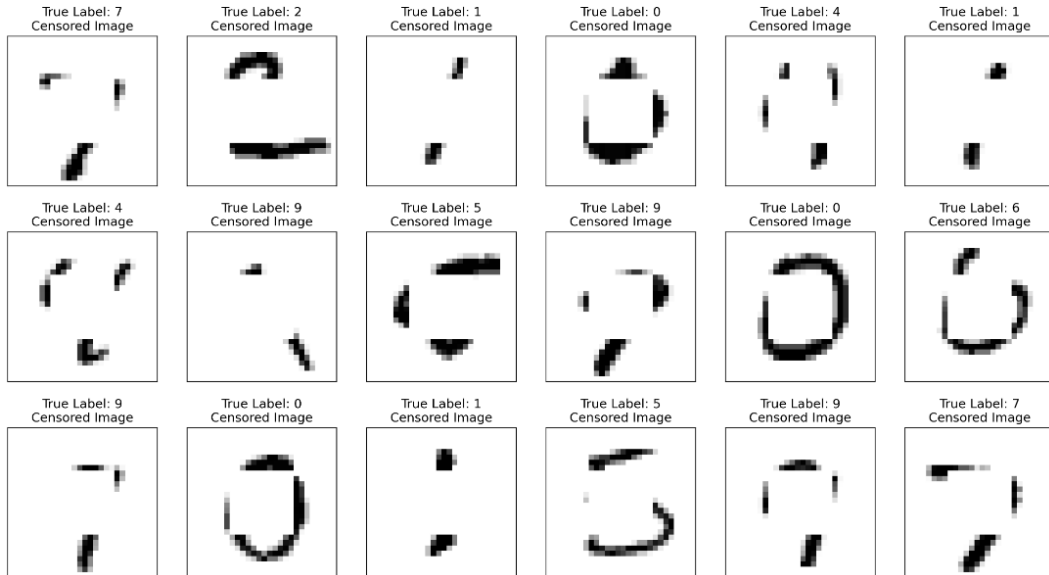


Figure 2: Examples of censored test images used during inference with missing pixels.

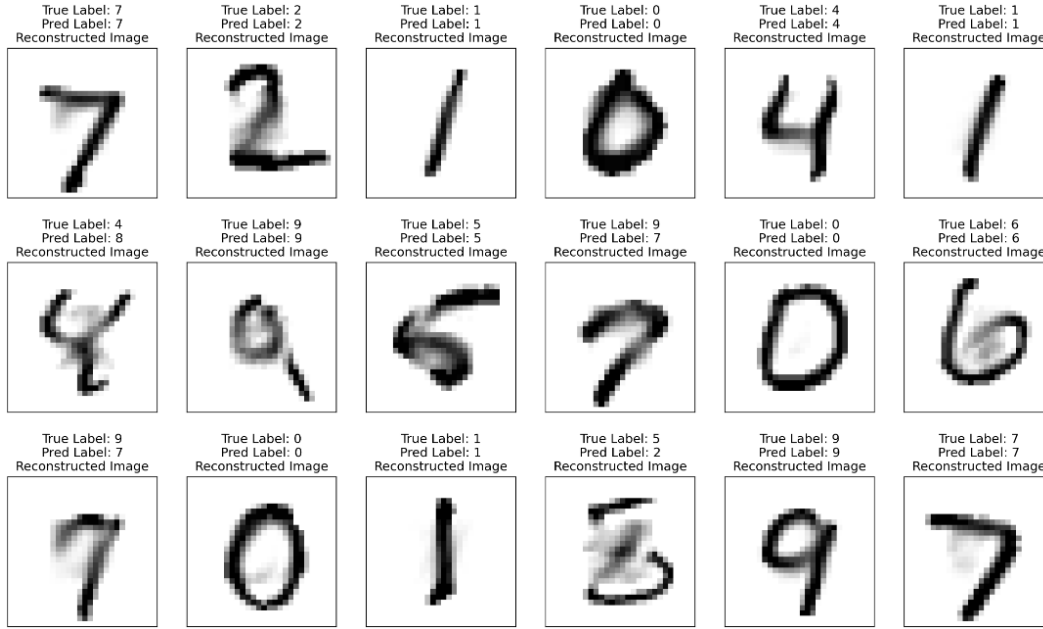


Figure 3: Reconstructed images obtained using the Gaussian conditional model after classification. The predicted labels and reconstructed digits are shown for each censored input.

3 Question 3: Experiments, Plotting and Analysis

3.1 Objective

The objective of this experiment is to compare the performance of two inference procedures:

1. Two-step inference (classification followed by reconstruction)
2. Single-step inference (joint classification and reconstruction)

The comparison is carried out under varying levels of missing pixels in the input images. The goal is to analyze how both methods behave in terms of:

- classification accuracy,
- inference time,
- reconstruction quality.

3.2 Experimental Setup

We use the MNIST dataset and a class-conditional Gaussian model trained as described in the lecture. Importantly, the training procedure is not modified. Only the inference stage is changed to compare the two methods.

To simulate missing data, a central square region of each image is censored (set to zero). The following censoring levels are used:

$$X[:, 4 : 24, 4 : 24] = 0 \quad (51\% \text{ missing})$$

$$X[:, 6 : 22, 6 : 22] = 0 \quad (32\% \text{ missing})$$

$$X[:, 8 : 20, 8 : 20] = 0 \quad (18\% \text{ missing})$$

$$X[:, 10 : 18, 10 : 18] = 0 \quad (8\% \text{ missing})$$

$$X[:, 12 : 16, 12 : 16] = 0 \quad (2\% \text{ missing})$$

These settings progressively reduce the amount of missing information, allowing us to observe how both inference procedures behave across different levels of data availability.

3.3 Evaluation Metrics

The following metrics are used for evaluation:

1. **Classification Accuracy:** Measures the proportion of correctly predicted class labels.
2. **Inference Time:** Measures the time required to perform prediction (classification stage only).
3. **Reconstruction Quality:** Evaluated qualitatively by visual inspection of reconstructed images. Additionally, missing-pixel Mean Squared Error (MSE) is used as supporting evidence.

3.4 Quantitative Results

Table 2: Comparison of two-step and single-step inference across censoring levels

Missing (%)	Acc (Two-step)	Acc (Single-step)	Time (Two-step, s)	Time (Single-step, s)
51	0.4070	0.2693	0.5823	0.5352
32	0.6976	0.5970	0.9846	1.1324
18	0.7857	0.7667	1.2062	1.2589
8	0.8315	0.8384	1.4705	1.5629
2	0.8525	0.8569	1.6731	1.7267

3.5 Plots

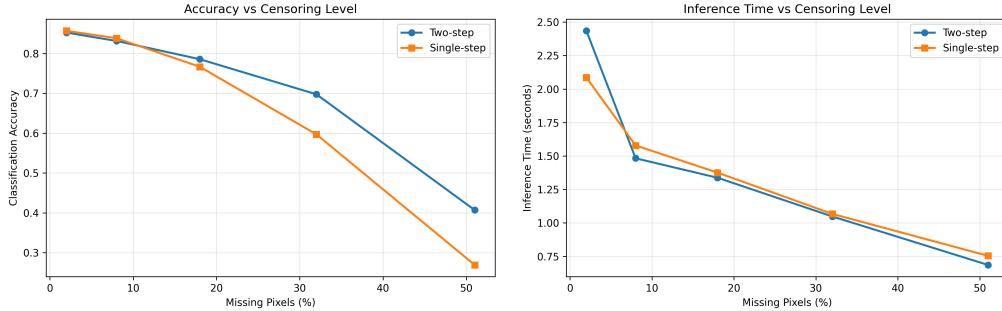


Figure 4: Left: Classification accuracy vs censoring level. Right: Inference time vs censoring level.

3.6 Analysis

The results clearly show that both inference methods are significantly affected by the amount of missing data. As the percentage of missing pixels decreases from 51% to 2%, the classification accuracy improves steadily for both methods. This behavior is expected, as a larger number of observed pixels provides more reliable information for estimating class probabilities.

At high censoring levels (51%, 32%, and 18%), the two-step method performs noticeably better than the single-step method. This is because the two-step method relies only on observed pixels for classification and does not attempt to jointly model missing values. As a result, it is more robust when the available information is very limited.

In contrast, the single-step method incorporates the uncertainty associated with missing pixels through the conditional covariance term. While this theoretically provides a more principled approach, it becomes sensitive to estimation errors when the number of observed pixels is small. This explains the lower accuracy of the single-step method under severe censoring.

As the censoring level decreases (8% and 2%), the performance gap between the two methods reduces, and the single-step method slightly outperforms the two-step method. In these cases, sufficient observed data is available, allowing the single-step method to effectively leverage correlations between observed and missing pixels.

Regarding inference time, both methods show an increasing trend as the censoring decreases. This is because more observed dimensions result in higher computational cost for evaluating Gaussian likelihoods. The single-step method is generally slightly slower due to the additional computations required for modeling conditional uncertainty.

For reconstruction quality, images reconstructed under heavy censoring appear blurred and incomplete for both methods. As the amount of missing data decreases, reconstructions become sharper and more visually consistent with the original digit structure. The missing-pixel MSE follows a similar trend, decreasing as more pixels are observed.

3.7 Conclusion

In summary, the two-step and single-step inference procedures exhibit different strengths depending on the level of missing data. The two-step method is more robust in high-missing scenarios, while the single-step method becomes advantageous when sufficient observed information is available.

These results confirm that although single-step inference is theoretically more expressive, its practical performance depends heavily on the quality and quantity of observed data. Therefore, the two methods are not equivalent and produce different behaviors across varying censoring levels.